

Assignment – Java Web Application Development

Module: Object-Oriented Programming / Web Application Development

Level: Year 2 Undergraduate

1. Assignment Description

Students are required to design and develop a dynamic web application using Java technologies.

The project must demonstrate a clear understanding of:

- Server-side programming
- Database integration
- Form handling
- Object-oriented concepts
- Web application architecture

2. What Is Expected from Students

Students must:

- Develop a functional Java-based web application using **Servlets and JSP** (or an equivalent approved Java web technology).
- Implement proper **user input handling** and **server-side validation**.
- Integrate a **relational database** (e.g., MySQL) for data storage and retrieval.
- Apply **MVC (Model-View-Controller)** design principles.
- Demonstrate the use of **object-oriented programming concepts** such as:
 - Classes
 - Inheritance
 - Encapsulation
- Ensure proper **session management**, where required.
- Design a **clean, user-friendly front-end interface**.
- Handle exceptions and errors appropriately.
- Organize the project structure according to the **standard Java web application format**.

3. Technical Requirements

- Use **Java (Servlets/JSP or an approved framework)**.
- Use a **relational database** for backend storage.
- Deploy and test the application using a web server such as **Apache Tomcat**.
- Follow proper **package structure and naming conventions**.
- Use **Prepared Statements** to prevent SQL injection.

4. Rules and Submission Guidelines

- The project must be developed **individually**, unless otherwise specified.
- Copying complete projects from the internet is **strictly prohibited**.
- Code must be **well-commented and properly formatted**.
- Submit:
 - Source code files
 - Database script file
- Include a short **project report** explaining:
 - System design
 - System functionality
- Ensure the application runs correctly before submission.

5. Evaluation Criteria

Marks will be awarded based on:

- Correct implementation of Java web technologies
- Database integration and query handling
- Application functionality and business logic
- Code quality and project structure
- User interface design and usability